

Track 24: Drone Programming

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Drone Basics → Flight Control → PX4/ArduPilot → MAVLink → Autonomous Missions → Computer Vision

Related: [Track 23 — ROS2](#) | [Track 26 — Computer Vision](#)

Track Overview

Who this is for: Aerospace, robotics, and CS students interested in UAV software and autonomous flight.

Prerequisites: Python; basic physics; [ROS2](#) helpful for 3-month+ tiers.

Tools: PX4 SITL / ArduPilot SITL, Gazebo, QGroundControl, MAVSDK-Python, Mission Planner, drone simulator.

Safety: All flight labs in simulation until 6-month+ tier with certified outdoor flying authorization.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Drone software awareness	Simulated waypoint mission
1 Month	80–100	Junior drone programmer	Autonomous takeoff/land script
3 Months	250–300	Internship-ready	Full mission planner capstone
6 Months	500–600	UAV developer entry	Vision-guided drone sim project
9 Months	750–900	Advanced UAV engineer	Real-world mission system (where permitted)

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Drone anatomy, flight dynamics, regulations	Safety + regulations quiz
D3–4	PX4/ArduPilot setup, SITL simulation	Simulated hover flight
D5–6	MAVLink protocol, telemetry	Telemetry dashboard
D7–8	Waypoint missions, QGroundControl	5-waypoint mission in sim
D9–10	Python MAVSDK scripting	Automated mission script

1-Month Foundation

Week	Topics	Project
W1	SITL setup, manual vs autonomous modes	Stable hover in sim
W2	MAVSDK: takeoff, land, goto	Scripted flight sequence
W3	Geofencing, failsafe, RTL	Safety-configured mission
W4	Mission planning, logging, analysis	Survey grid mission in sim

3-Month Job Ready

Month	Topics	Projects
M1	PX4/ArduPilot deep dive, parameters	Tuned flight controller sim
M2	ROS2 + PX4 integration (optional)	Offboard control node
M3	Mission automation, capstone	Autonomous delivery/survey mission

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Computer vision: object tracking from drone	Follow-target in sim
M5	Swarm basics, multi-drone coordination	2-drone sim coordination
M6	Capstone	Vision-guided autonomous UAV system

9-Month Advanced

Capstones: Industrial inspection drone pipeline + swarm coordination platform | **Level:** L5

Regulatory note: Outdoor flying requires DGCA/local compliance; institution must obtain permissions.

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	Flight dynamics, MAVLink, safety
Lab Completion	40%	Sim missions complete successfully
Project	30%	Mission reliability, safety compliance
Portfolio	10%	Flight logs, videos, code repos

Appendix: Mentor Notes

Never fly without geofence and RTL configured. Simulation-only until student passes safety certification.